10.7.90

EE25BTECH11042 - Nipun Dasari

Question:
A circle touches the line $y = x$ at a point P such that $OP = 4\sqrt{2}$, where O is the origin. The circle contains the point $(-10, 2)$ in its interior and the length of its chord on the line $x + y = 0$ is $6\sqrt{2}$. Determine the equation of the circle.

Solution:

Let the point of tangency be $\mathbf{P}$. Since it lies on the line $y = x$, its coordinates must be equal.

$$\mathbf{P} = \begin{pmatrix} p \\ p \end{pmatrix} \tag{0.1}$$

The distance from the origin is given as $\|\mathbf{P}\| = 4\sqrt{2}$.

$$\|\mathbf{P}\| = \sqrt{p^2 + p^2} = |p|\sqrt{2} = 4\sqrt{2}$$
$$\implies |p| = 4 \tag{0.2}$$

This gives two possible points of tangency.

$$\mathbf{P}_1 = \begin{pmatrix} 4 \\ 4 \end{pmatrix} \quad \text{and} \quad \mathbf{P}_2 = \begin{pmatrix} -4 \\ -4 \end{pmatrix} \tag{0.3}$$

Let the center of the circle be $\mathbf{C}$.

$$\mathbf{C} = \begin{pmatrix} h \\ k \end{pmatrix} \tag{0.4}$$

The center lies on a line perpendicular to the tangent line, which has the matrix equation $\begin{pmatrix} 1 & -1 \end{pmatrix} \mathbf{x} = 0$. The normal to the tangent is $\mathbf{n}_t = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. A normal to the line of centers, $\mathbf{n}_c$, must be orthogonal to $\mathbf{n}_t$.

$$\mathbf{n}_c = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{0.5}$$

The equation for the line of centers is $\mathbf{n}_c^\top \mathbf{x} = d$, where $d = \mathbf{n}_c^\top \mathbf{P}$. For $\mathbf{P}_1$, the line of centers is:

$$\begin{pmatrix} 1 & 1 \end{pmatrix} \mathbf{C} = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 4 \\ 4 \end{pmatrix} = 8 \tag{0.6}$$

For $\mathbf{P}_2$, the line of centers is:

$$\begin{pmatrix} 1 & 1 \end{pmatrix} \mathbf{C} = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} -4 \\ -4 \end{pmatrix} = -8 \tag{0.7}$$

The circle has a chord on the line with equation $\begin{pmatrix}1 & 1\end{pmatrix}\mathbf{x} = 0$. The distance from the center $\mathbf{C}$ to this line is d_{chord}.

$$d_{chord} = \frac{|\mathbf{n}_c^\top \mathbf{C}|}{\|\mathbf{n}_c\|} = \frac{|h+k|}{\sqrt{2}} \tag{0.8}$$

The radius r, distance d_{chord}, and half the chord length ($L = 3\sqrt{2}$) are related by $r^2 = d_{chord}^2 + L^2$.

$$r^2 = \left(\frac{|h+k|}{\sqrt{2}}\right)^2 + (3\sqrt{2})^2 = \frac{(h+k)^2}{2} + 18 \tag{0.9}$$

The radius is also the distance from the center to the point of tangency, $r^2 = \|\mathbf{C} - \mathbf{P}\|^2$.

For the first case ($\begin{pmatrix}1 & 1\end{pmatrix}\mathbf{C} = 8$ and $\mathbf{P} = \begin{pmatrix}4\\4\end{pmatrix}$):

$$r^2 = \frac{8^2}{2} + 18 = 50 \tag{0.10}$$
$$\|\mathbf{C} - \mathbf{P}\|^2 = (h-4)^2 + (k-4)^2 = (h-4)^2 + (8-h-4)^2 = 2(h-4)^2 = 50$$
$$(h-4)^2 = 25 \implies h-4 = \pm 5 \implies h = 9 \text{ or } h = -1 \tag{0.11}$$

This gives centers $\mathbf{C}_A = \begin{pmatrix}9\\-1\end{pmatrix}$ and $\mathbf{C}_B = \begin{pmatrix}-1\\9\end{pmatrix}$.

For the second case ($\begin{pmatrix}1 & 1\end{pmatrix}\mathbf{C} = -8$ and $\mathbf{P} = \begin{pmatrix}-4\\-4\end{pmatrix}$):

$$r^2 = \frac{(-8)^2}{2} + 18 = 50 \tag{0.12}$$
$$\|\mathbf{C} - \mathbf{P}\|^2 = (h+4)^2 + (k+4)^2 = (h+4)^2 + (-8-h+4)^2 = 2(h+4)^2 = 50 \tag{0.13}$$
$$(h+4)^2 = 25 \implies h+4 = \pm 5 \implies h = 1 \text{ or } h = -9 \tag{0.14}$$

This gives centers $\mathbf{C}_C = \begin{pmatrix}1\\-9\end{pmatrix}$ and $\mathbf{C}_D = \begin{pmatrix}-9\\1\end{pmatrix}$.

The point $\mathbf{Q} = \begin{pmatrix}-10\\2\end{pmatrix}$ is in the circle's interior, so $\|\mathbf{Q} - \mathbf{C}\|^2 < r^2 = 50$.

$$\text{For } \mathbf{C}_A = \begin{pmatrix}9\\-1\end{pmatrix} : \|\begin{pmatrix}-19\\3\end{pmatrix}\|^2 = 370 > 50 \tag{0.15}$$
$$\text{For } \mathbf{C}_B = \begin{pmatrix}-1\\9\end{pmatrix} : \|\begin{pmatrix}-9\\-7\end{pmatrix}\|^2 = 130 > 50 \tag{0.16}$$
$$\text{For } \mathbf{C}_C = \begin{pmatrix}1\\-9\end{pmatrix} : \|\begin{pmatrix}-11\\11\end{pmatrix}\|^2 = 242 > 50 \tag{0.17}$$
$$\text{For } \mathbf{C}_D = \begin{pmatrix}-9\\1\end{pmatrix} : \|\begin{pmatrix}-1\\1\end{pmatrix}\|^2 = 2 < 50 \tag{0.18}$$

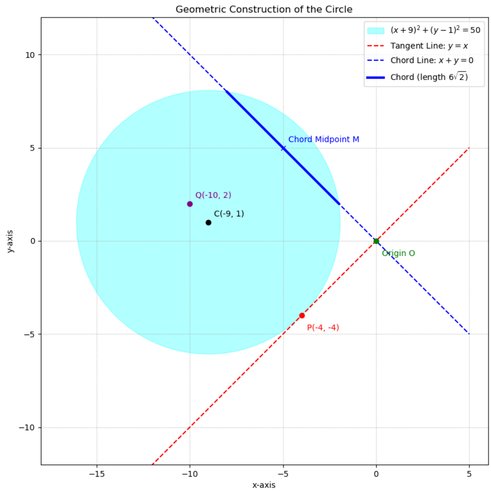
The correct center is $\mathbf{C}_D = \begin{pmatrix}-9\\1\end{pmatrix}$.

The center is $\begin{pmatrix}-9\\1\end{pmatrix}$ and radius squared is $r^2 = 50$. The equation is $(x+9)^2+(y-1)^2 = 50$.

$$x^2 + y^2 + 18x - 2y + 32 = 0 \tag{0.19}$$

In the matrix form $\mathbf{x}^\top \mathbf{V}\mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$:

$$\begin{pmatrix}x & y\end{pmatrix}\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} + 2\begin{pmatrix}9 & -1\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} + 32 = 0 \tag{0.20}$$



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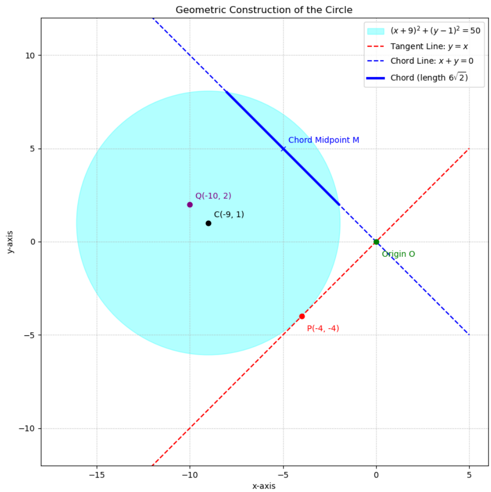
Geometric Construction of the Circle
$(x+9)^2+(y-1)^2=50$
Tangent Line: $y=x$
Chord Line: $x+y=0$
Chord (length $6\sqrt{2}$)
Chord Midpoint M
Q(-10, 2)
C(-9, 1)
Origin O
P(-4, -4)
10
5
0
−5
−10
y-axis
−15
−10
−5
0
5
x-axis

Fig